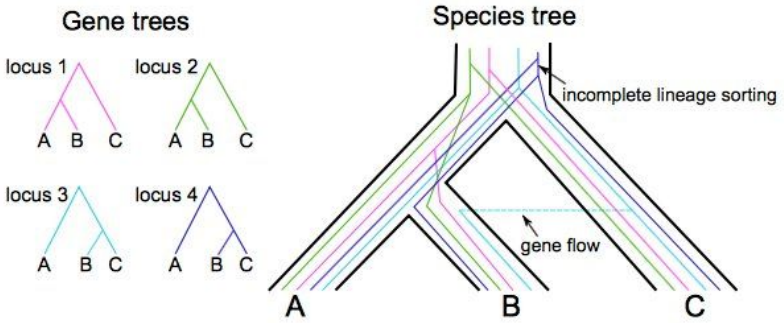
BME 105: Genetics in the Genomics Era

Lecture 23: Intro to phylogenetic inference



Instructor: Dr. Shelbi Russell,
Assistant Professor

Course website on canvas.ucsc.edu

Lecture 23 Learning Objectives

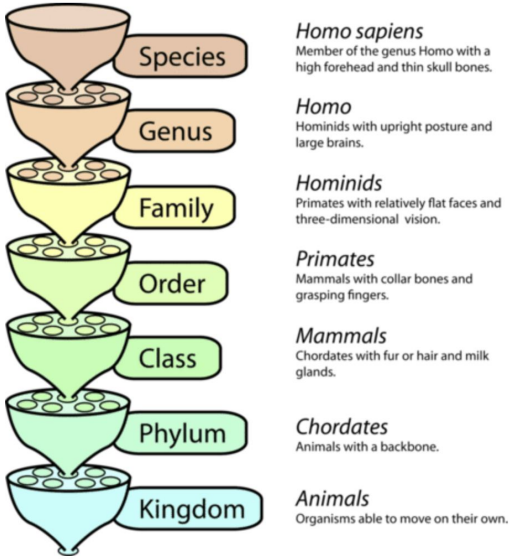
- Be able to read and interpret phylogenetic trees. Know the words used to describe them.
- Understand why phylogenetic inference is reliant on shared and derived character states.
- Know what types of characters (e.g., DNA vs morphology) can be used for phylogenetic inference.
- Be able to describe why monophyletic clades are inferred through the identification of homologous character states.
- Understand why heuristics are necessary in phylogenetic inference (an np-hard problem), and know the optimality criteria used to find the "best" tree (distance, parsimony, maximum likelihood)
- Understand how uncertainty in phylogenetic inference is represented, e.g., with bootstrap analysis values.
- Know what molecular clocks are and how they are used to date major biological events
- Understand why many loci and sites are needed to infer well-resolved phylogenies: nucleotide substitutions saturate, recombination unlinks locus histories, constraint limits phylogenetic signal, selected loci may obscure demography, etc.
- Be able to articulate the difference between a gene tree and a species tree

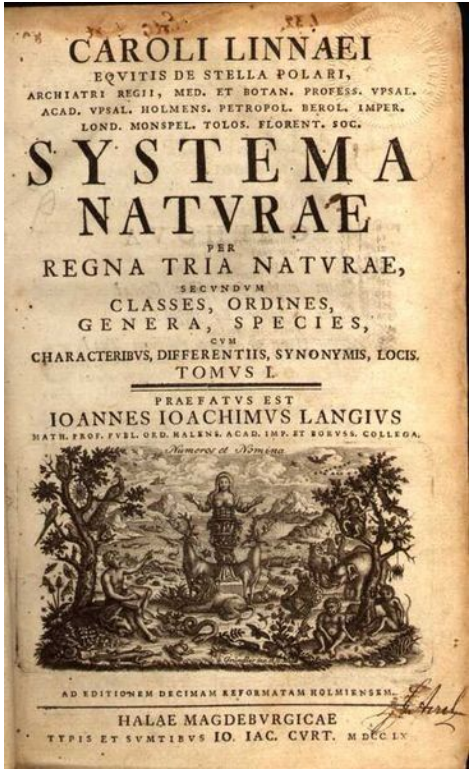
Biological organisms exhibit grouped similarities that have been observed for centuries

Aristotle (384-322 BC) introduced classification of organisms by type and binomial definition.

Carl (Carolus) Linnaeus (1707-1778) formalized our current taxonomic system ~ ***Linnaean taxonomy***

- Binomial nomenclature
- Classes, orders, genera and species, with characteristics, differences, synonyms, places.
  - Kingdoms have been replaced with Domains
- ~100 years before Mendel

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EQVITIS DE STELLA POLARI,
ARCHIATRI REGII, MED. ET BOTAN. PROFESS. VPSAL.
ACAD. VPSAL. HOLMENS. PETROPOL. BEROL. IMPER.
LOND. MONSPEL. TOLOS. FLORENT. SOC.

SYSTEMA NATVRAE
PER
REGNA TRIA NATVRAE,
SECVNDVM
CLASSES, ORDINES,
GENERA, SPECIES,
CVM
CHARACTERIBVS, DIFFERENTIIS, SYNONYMIS, LOCIS.
TOMVS I.

PRAEFATVS EST
IOANNES IOACHIMVS LANGIVS
MATH. PROF. PVBL. ORD. HALENS. ACAD. IMP. ET BORVSS. COLLEGA.

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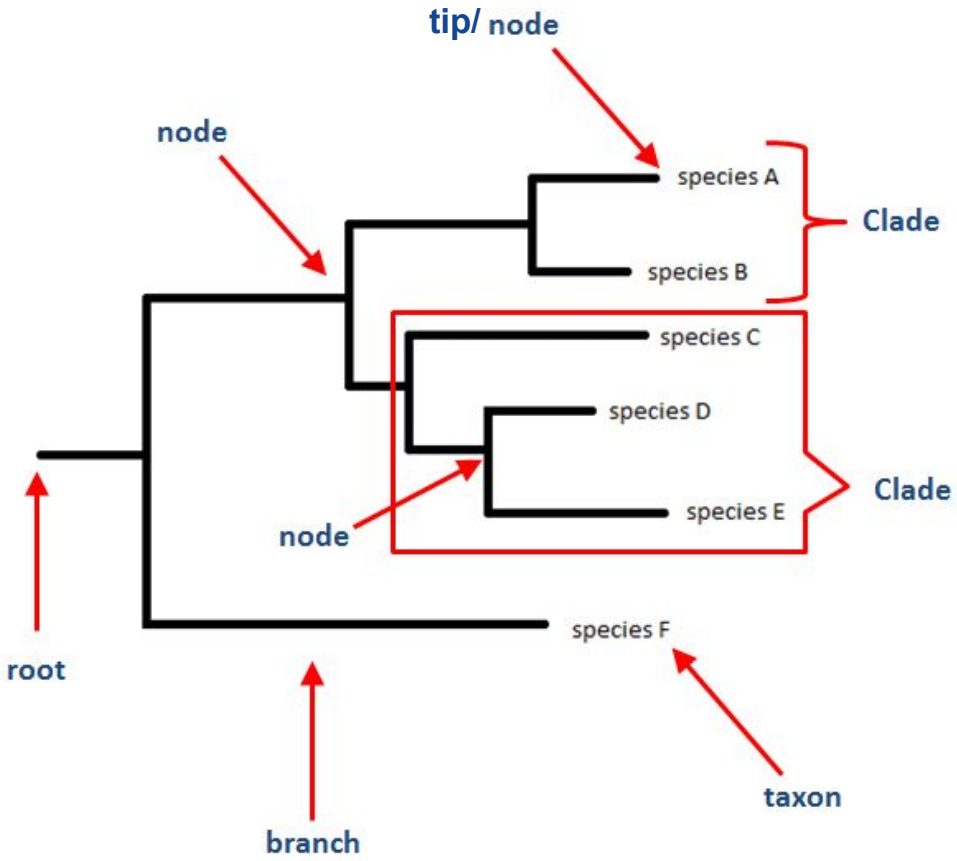
HALAE MAGDEBVRGICAE
TYPIS ET SVMTIBVS IO. IAC. CVRT. MDCCLX

Reading a Phylogenetic Tree



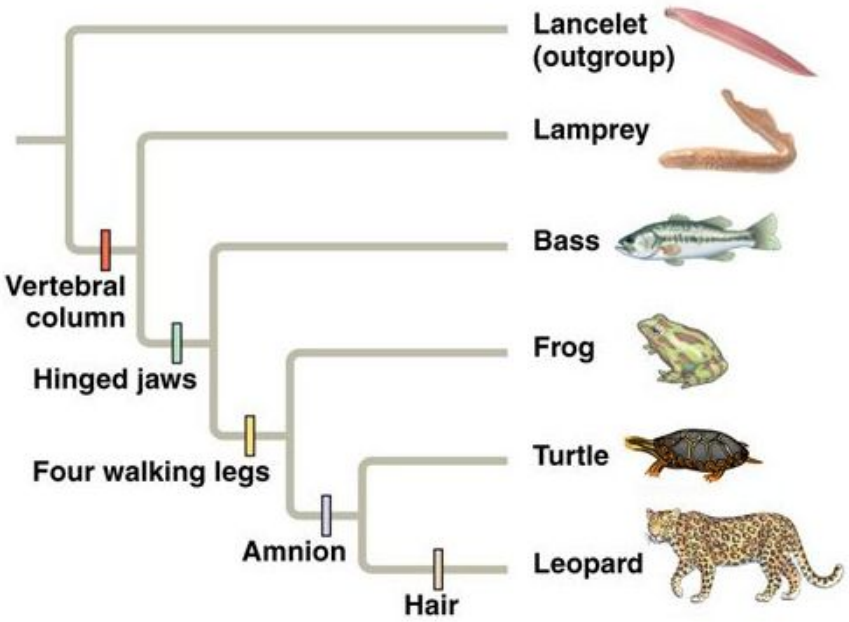
Slide from Russ Corbett-Detig

Phylogenies are built on shared, derived character states

- Characters include:
  - **Morphology**
  - Behavior
  - DNA/proteins

Homology == character similarity due to shared ancestry

CHARACTERS \ TAXA	Lancelet (outgroup)	Lamprey	Bass	Frog	Turtle	Leopard
Vertebral column (backbone)	0	1	1	1	1	1
Hinged jaws	0	0	1	1	1	1
Four walking legs	0	0	0	1	1	1
Amnion	0	0	0	0	1	1
Hair	0	0	0	0	0	1



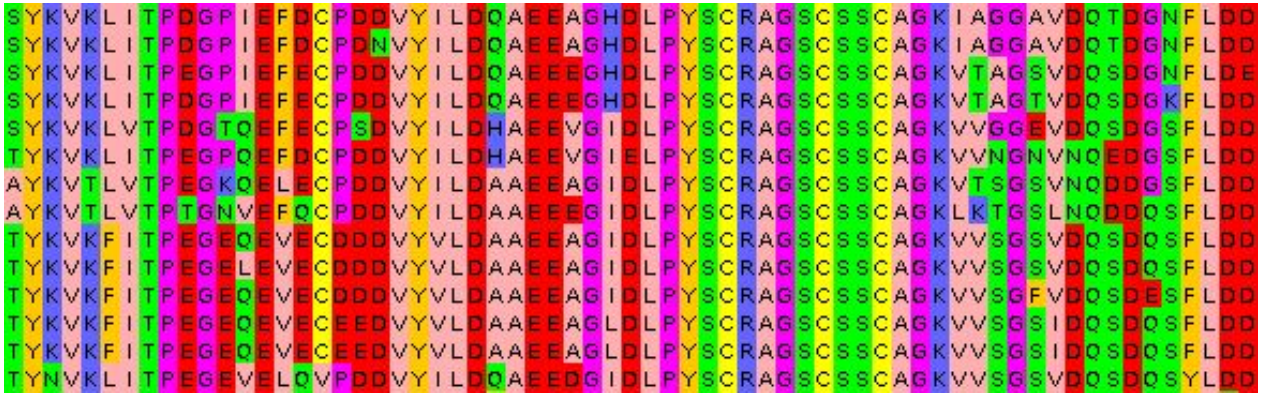
Phylogenies are built on shared, derived character states

- Characters include:
  - Morphology
  - Behavior
  - **DNA/proteins**

Homology == character similarity due to shared ancestry

Each amino acid (or encoding nucleotide) is a character

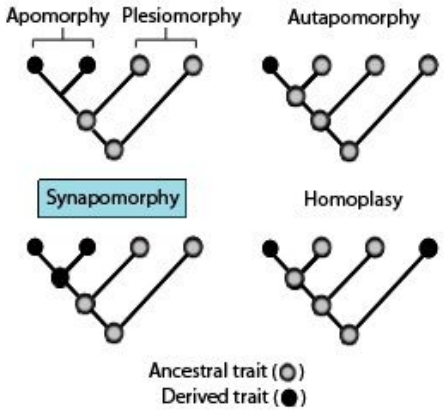
A multi-sequence alignment is a statement about homology



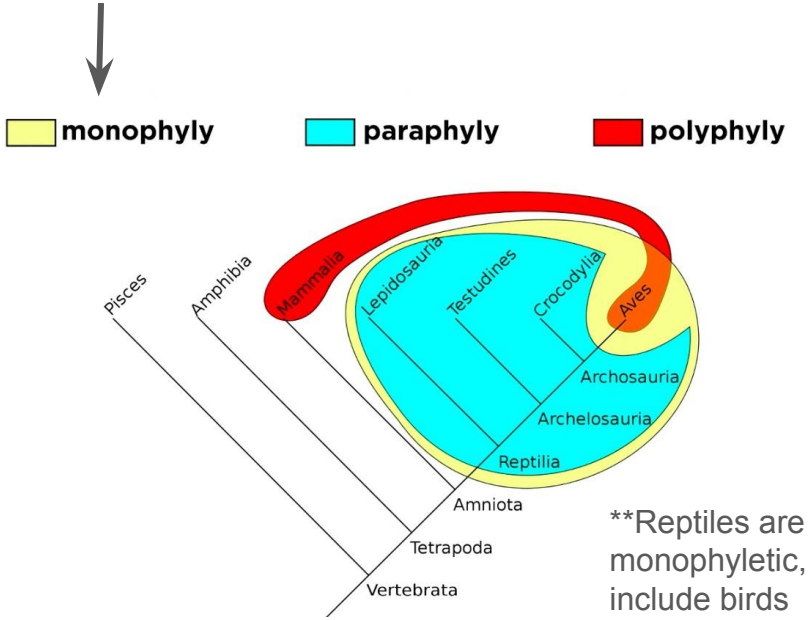
Phylogenies are built on shared, derived character states

- Characters include:
  - Morphology
  - Behavior
  - DNA/proteins

- Only shared (i.e., homologous) <u>&</u> <u>derived</u> traits are informative:

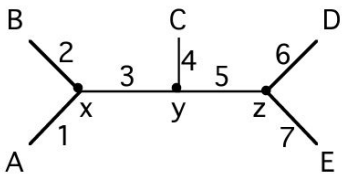


Synapomorphies inform on monophyletic relationships



**Reptiles are only monophyletic, if you include birds

Tree rooting implies directionality (ancestor-descendent relationships)



. Branches: If there are n-taxa in an unrooted tree, there are $2n$-3 branches, labeled 1 – 7.

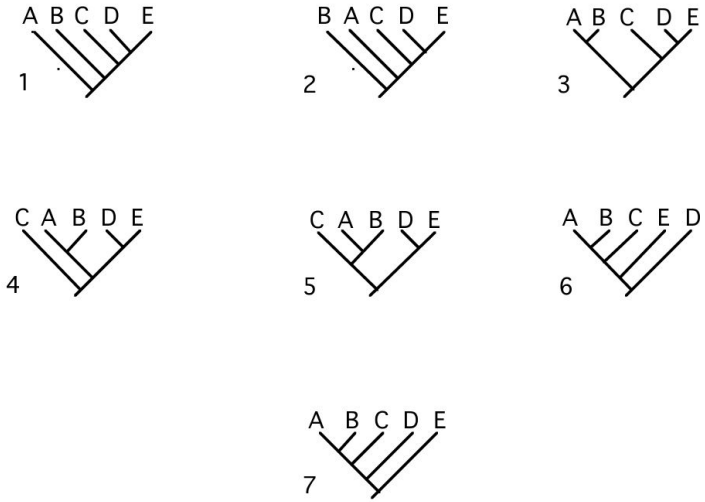
- 1, 2, 4, 6, & 7 are external, or terminal branches (leaves in math jargon)
- 3 & 5 are internal branches (edges); these represent splits, or bipartitions of taxa.

2. Nodes are points where branches intersect or terminate.

Nodes A – E are terminals, or terminal nodes (there are n), whereas nodes x, y, & z are internal nodes (there are n-2 internal nodes in an unrooted tree).

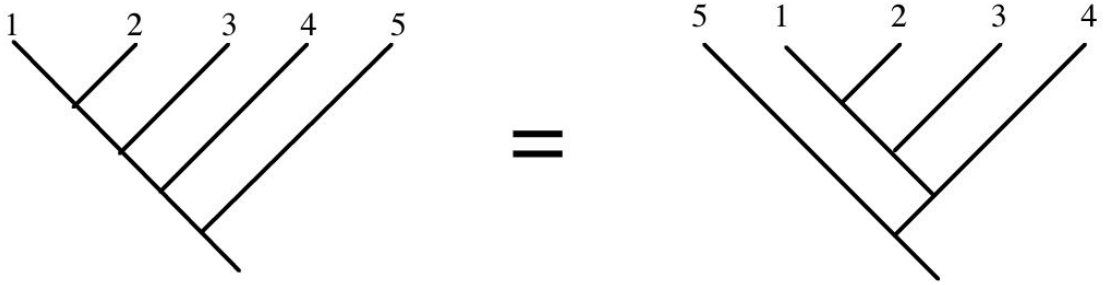
Trees can be represented in parenthetical notation (Newick format): ((A,B),C,(D,E))

All of the rooted trees below can be derived from the unrooted tree by placing the root (common ancestor) on each branch. So, there are $2n$-3 possible ways to root the tree (10 - 3=7 for our 5-taxon example) and these rooted trees *do* imply directionality; nodes deeper in the tree represent divergence events that happened earlier. Furthermore, this rooting exercise illustrates how rapidly the number of alternative trees grows as one adds taxa. Addition of a single node (the root node) resulted in the seven rooted trees below being derivable from the single unrooted tree above.



While rooting changes the relationships among taxa, rotating branches around nodes does not

note that there is free rotation around nodes:



Possible tree space is vast, and searching it is impossible

The number of possible trees for N taxa is given by:

$$B(N) = \prod_{i=3}^{N} (2i - 5)$$

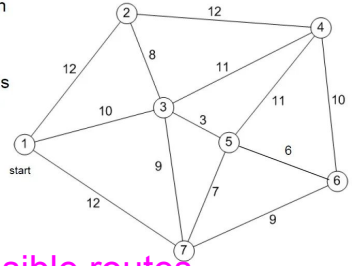
Number of Taxa	Number of Unrooted Trees
3	1
4	3
5	15
6	105
7	945
8	10,395
9	135,135
10	2.027×10^{6}
22	3×10^{23}
50	3×10^{74}
100	2×10^{82}
1000	$2 \times 10^{2,860}$
10,000,000	$5 \times 10^{68,667,340}$

So, the task of phylogeny inference is indeed formidable, which is why the idea that phylogenetic trees can be "*made easy*" (Hall, B. 2017. *Phylogenetic Trees Made Easy*, Oxford University Press, Oxford) is nonsensical.

A NP-hard problem:

The Traveling Salesman Problem

- Starting from city 1, the salesman must travel to all cities once before returning home
- The distance between each city is given, and is assumed to be the same in both directions
- Only the links shown are to be used
- Objective - Minimize the total distance to be travelled



n cities, n!/2 = possible routes

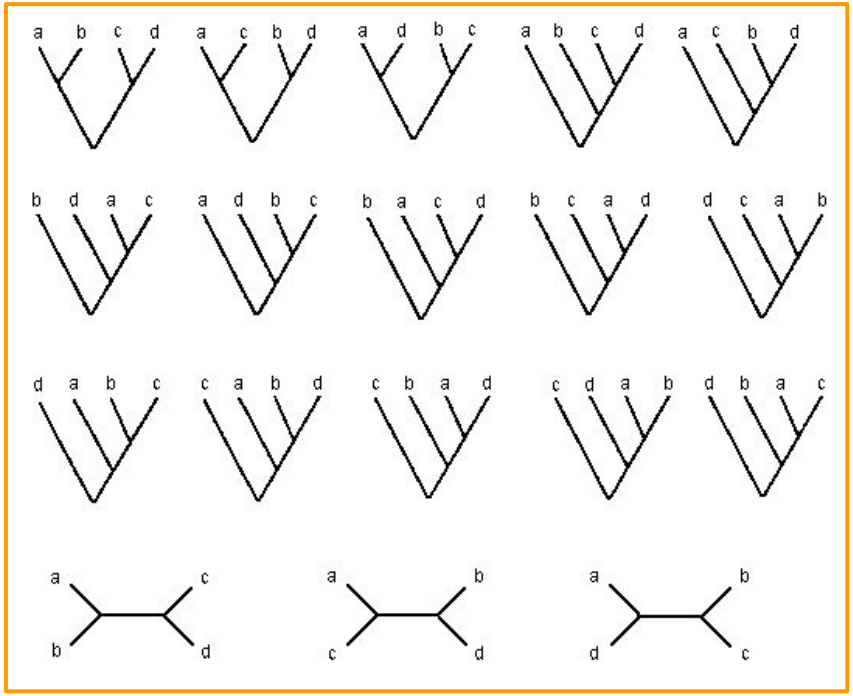
E.g., 31 cities = 1.3×10^{32} routes

Number of rooted trees grows even faster!

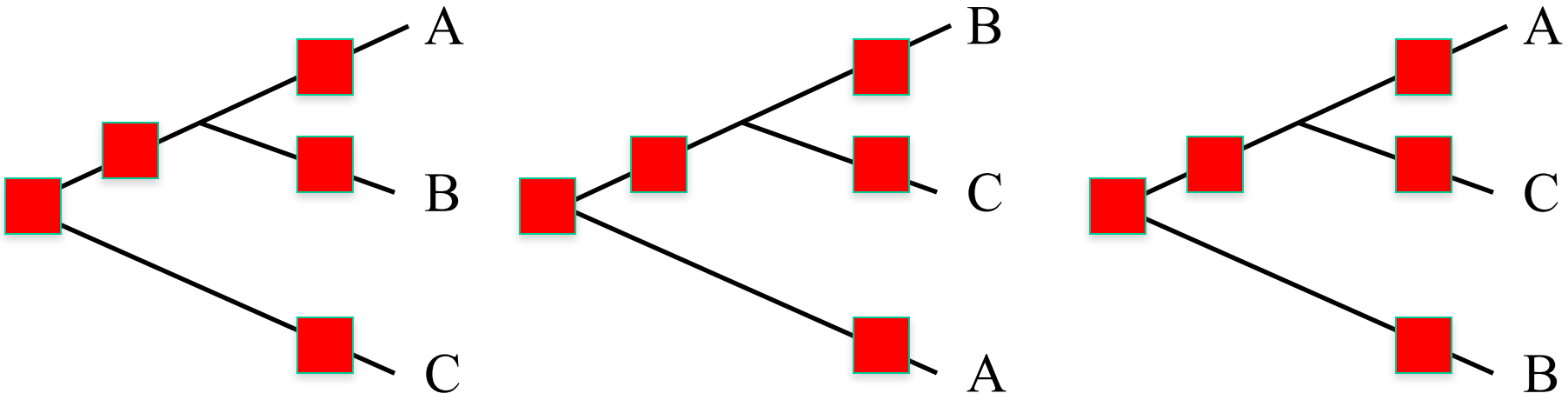
rooted trees =
(2n−3)! / 2^(n−2) * (n−2)!

unrooted trees =
(2n−5)! / 2^(n−3) * (n−3)!

# of species	# of rooted trees (R)	# of unrooted trees (U)
4	15	3
5	105	15
6	945	105
7	10,395	945
8	135,135	10,395
9	2,027,025	135,135
10	34,459,425	2,027,025
11	654,729,075	34,459,425
12	13,749,310,575	654,729,075



If there are four taxa, 15 possible trees!



Slide from Russ Corbett-Detig

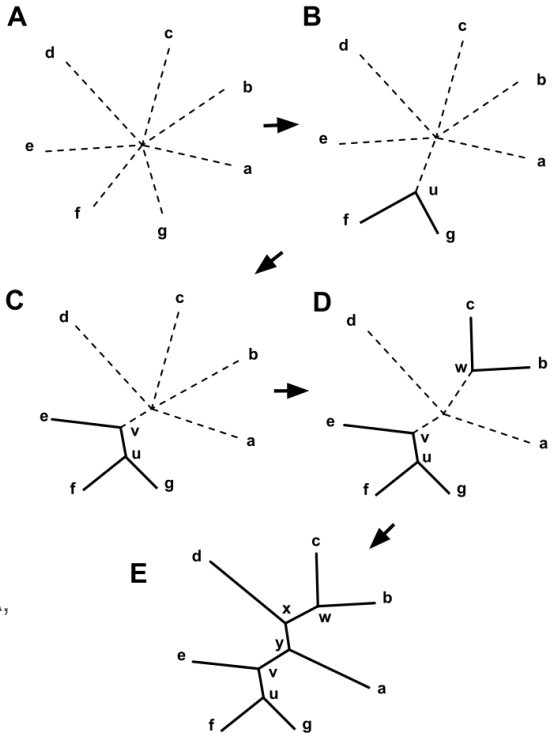
How do we find the “true” tree among so many possibilities?

Heuristics - approaches employ practical methods that are not guaranteed to be optimal, perfect, or rational, but are nevertheless sufficient for reaching an immediate, short-term goal or approximation

Optimality criteria - how we define ideal trees

Shortest (sequence) distance → **Distance methods** (UPGMA, Neighbor-joining, etc.)

** These trees are often made to serve as the starting trees for ML and Bayesian approaches



Starting with a star tree (A), the Q matrix is calculated and used to choose a pair of nodes for joining, in this case f and g. These are joined to a newly created node, u, as shown in (B). The part of the tree shown as solid lines is now fixed and will not be changed in subsequent joining steps. The distances from node u to the nodes a-e are computed from equation (3). This process is then repeated, using a matrix of just the distances between the nodes, a,b,c,d,e, and u, and a Q matrix derived from it. In this case u and e are joined to the newly created v, as shown in (C). Two more iterations lead first to (D), and then to (E), at which point the algorithm is done, as the tree is fully resolved.

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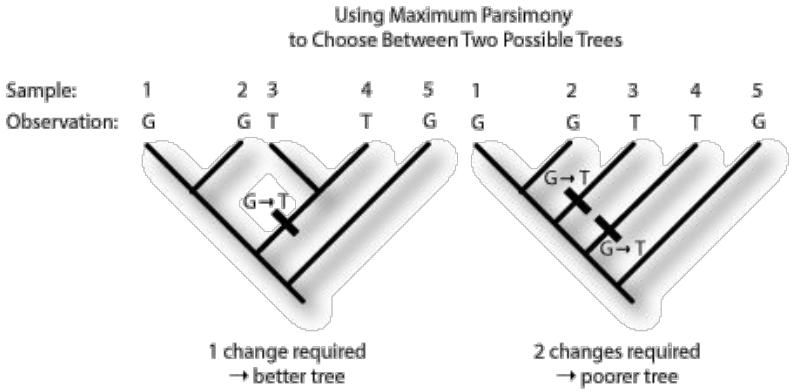
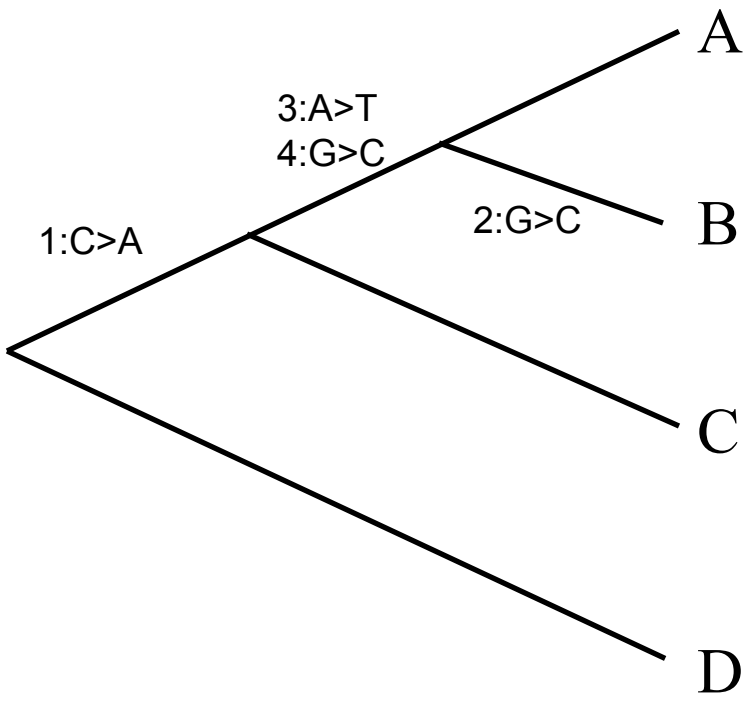
Shortest steps → **Maximum parsimony**



Simple Tree-Building Algorithms: Step-Wise Parsimony

Taxon	1	2	3	4
A	A	G	T	C
B	A	C	T	C
C	A	G	A	G
D	C	G	A	G



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Shortest steps → **Maximum parsimony**

Most likely tree, given the data → **Maximum likelihood**

ML objective is to maximize*:*
P(tree|data)

Iterate over possible trees, calculating P(data|tree)

Keep tree with maximum P(data|tree)

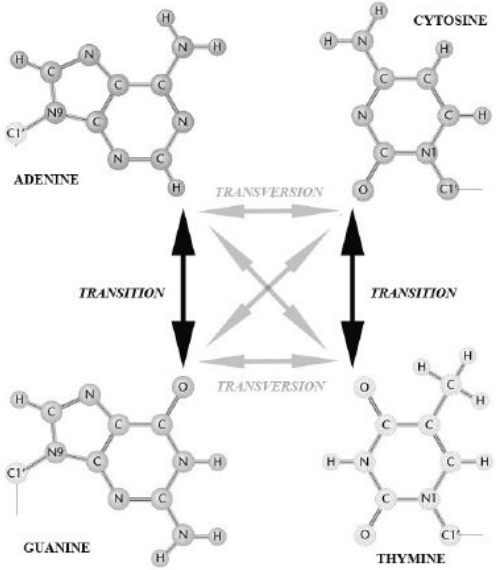
Assumptions and evolutionary models

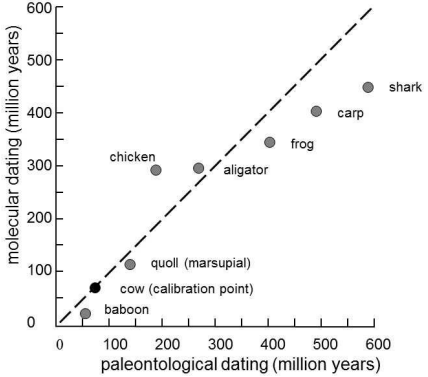
- Characters evolve independently: state changes are not correlated

- Substitution models:
  - Time-reversible (C->G and G->C occur at the same rate)
  - Transitions more common than transversions
  - Can test model fit with likelihood ratio test

- Outgroup taxa polarize substitutions
  - Assumption about evolutionary relationships
  - A type of tree topology prior
  - Alternative option: midpoint rooting

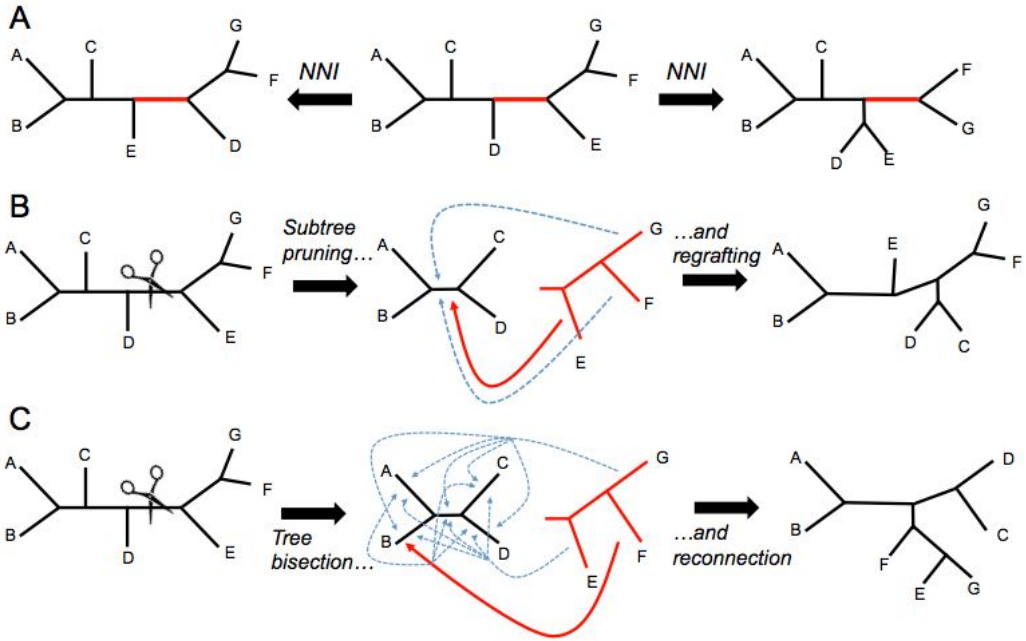
- Branch lengths ~ substitution rates





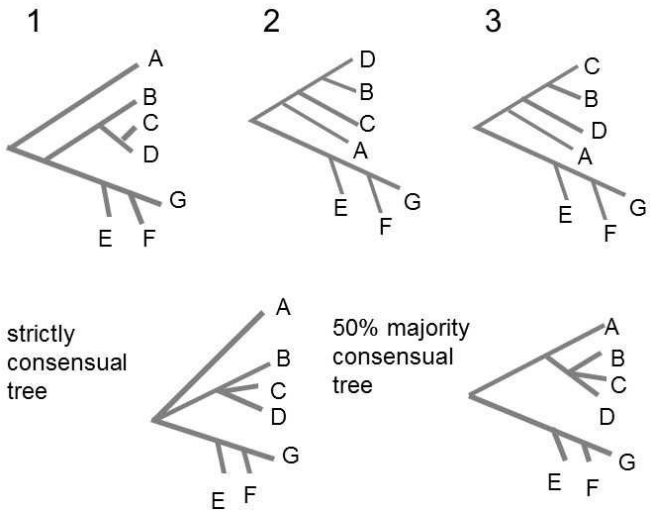
Optimal tree search space is huge, requiring approaches for finding optimal topologies

- Start with a neighbor-joining tree that doesn't do "too bad" of a job explaining the data
- Explore possible improvements in the topology:
  - Nearest neighbor interchanges
  - Subtree pruning and regrafting
  - Tree bisection and reconnection



Measuring statistical support for phylogenies

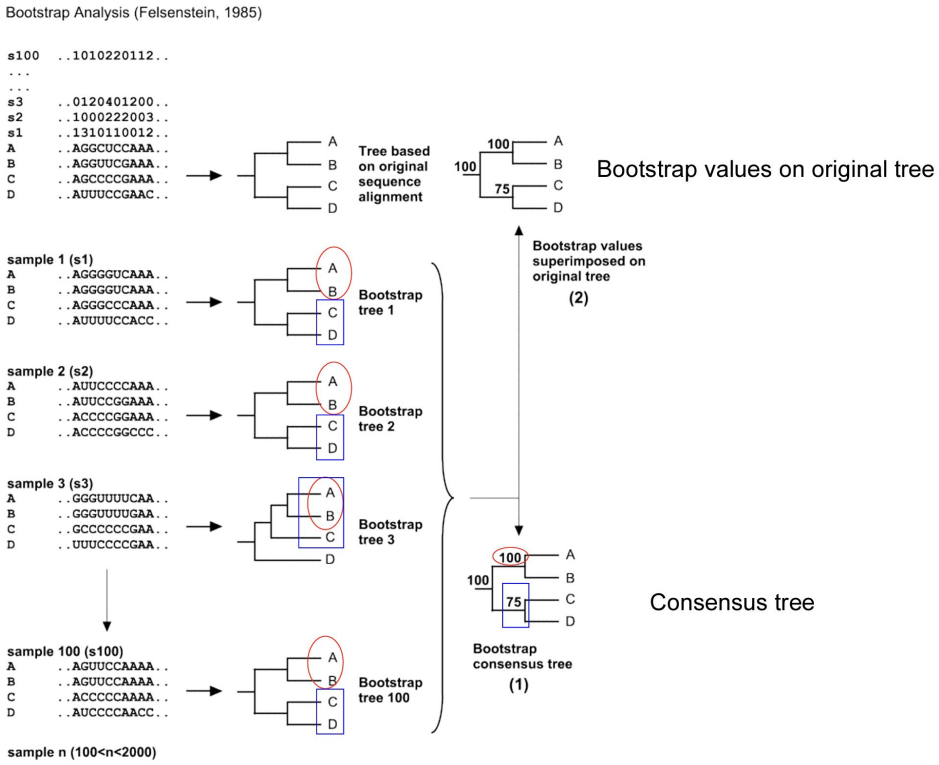
- Final trees can be:
  - Maximum likelihood/parsimony trees
  - Consensus tree from multiple equally probable trees
  - Multispecies coalescent (described later)



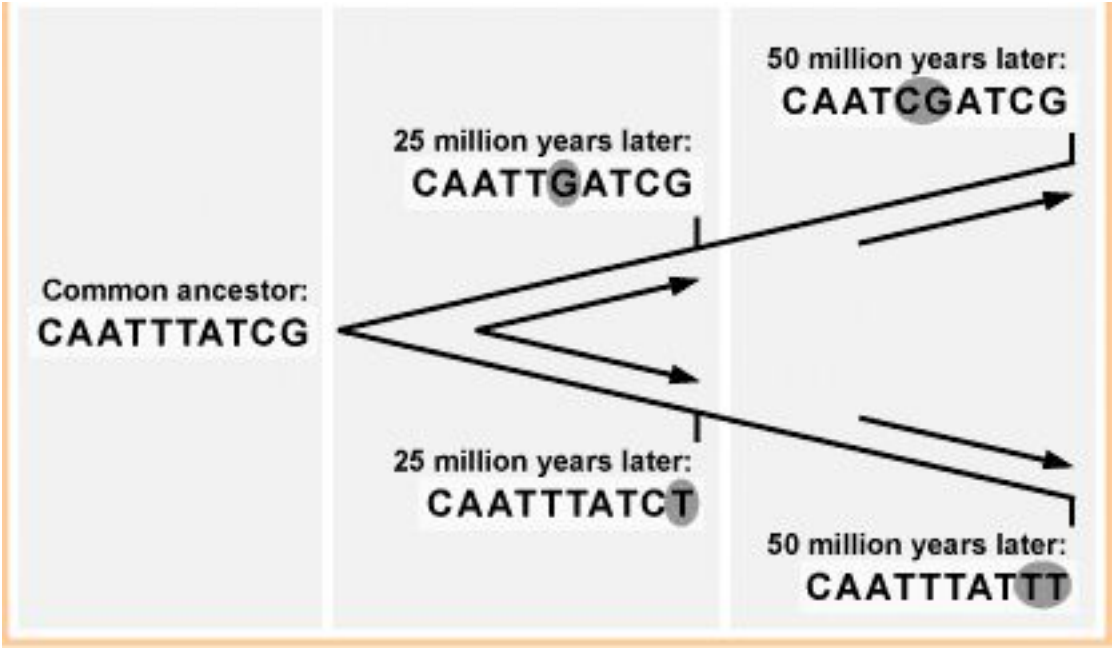
Measuring statistical support for phylogenies

- Final trees can be:
  - Maximum likelihood trees
  - Consensus tree
  - Multispecies coalescent

- Support for the final tree can be measured from the variance among topologies:
  - **Bootstrapping** - resampling sites/characters with replacement
    - Run 100-1000 resampling runs
    - Cutoffs >50% of bootstrap trees matched ML tree topology
  - **Jackknifing** - resampling sites/characters without replacement
  - **Bayesian posterior probabilities** - probability of the data, given the tree



Molecular Clock to Infer Timing of Major Speciation Events

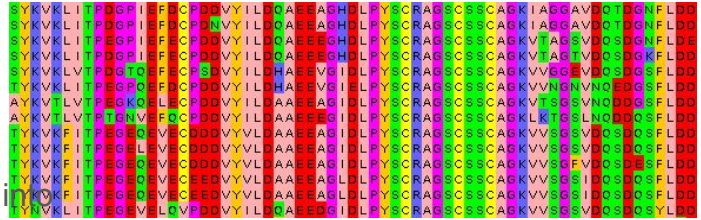


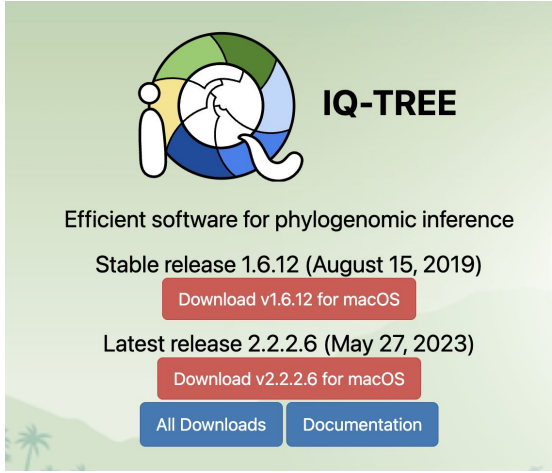
100 million years between two extant taxa

Slide from Russ Corbett-Detig

How do you make a genealogy or phylogeny?

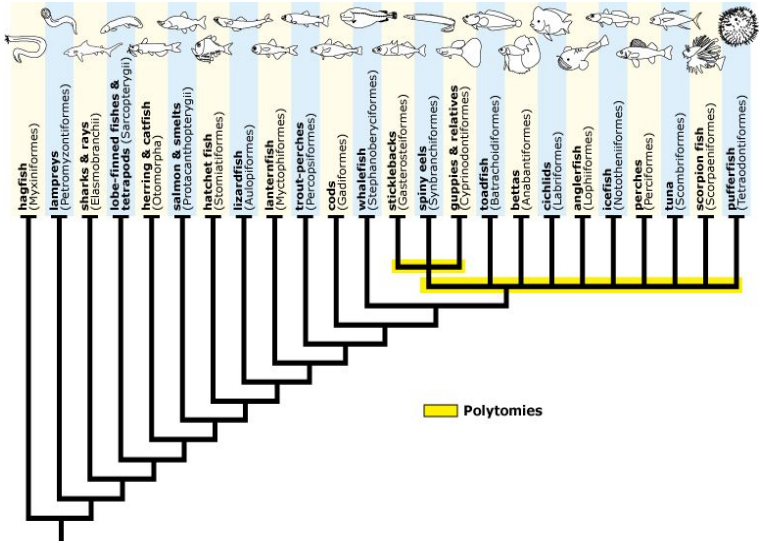
1. Collect a set of homologous sequences and/or characters
    a. Make your data or find it by BLAST in NCBI/EMBL
    b. Choose a set of **out-group** sequences to root your tree
2. Align sequences in a multi-sequence alignment and code characters in a matrix
    a. Gene alignment in e.g., MAFFT, Clustal, MUSCLE, etc.
3. Pick an inference + statistical support approach
    a. Parsimony works well if you don't need to model missing substitutions
    b. Maximum likelihood + bootstraps is generally a great choice (*e.g.*, IQ-Tree or RAxML (a bit old)).
    c. Bayesian approaches are great (e.g., MrBayes) and output posterior probabilities, but take too long to run on big datasets and are often overkill (ML produces similar results).

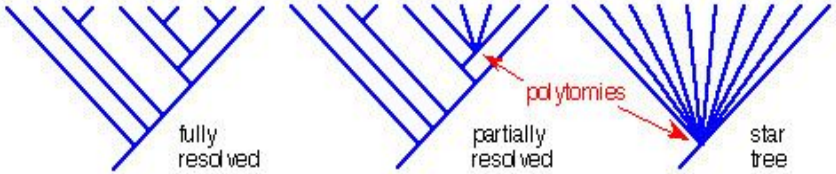




Gene tree inference vs species tree inference

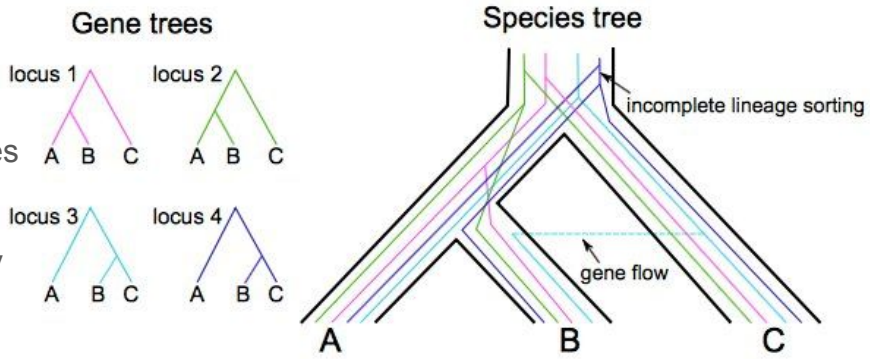
- All extant organisms had ancestors
- The relationships among all extant groups can be depicted through the patterns of divergence from their shared ancestors
- This generally takes the form of a bifurcating tree
  - True (“hard”) polytomies can happen, if radiation/diversification was fast.
  - Artifactual (“soft”) polytomies result from poor data resolution.





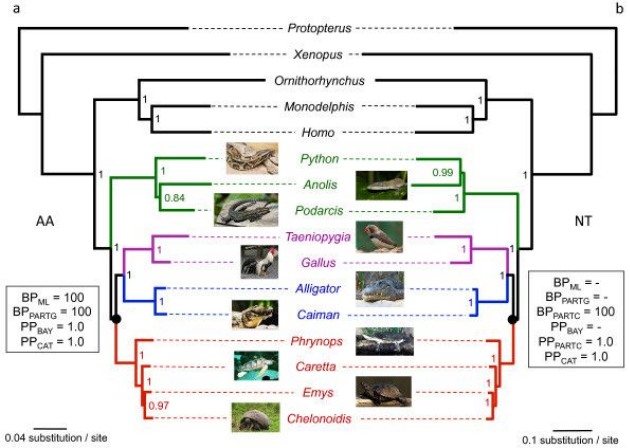
Gene tree inference vs species tree inference

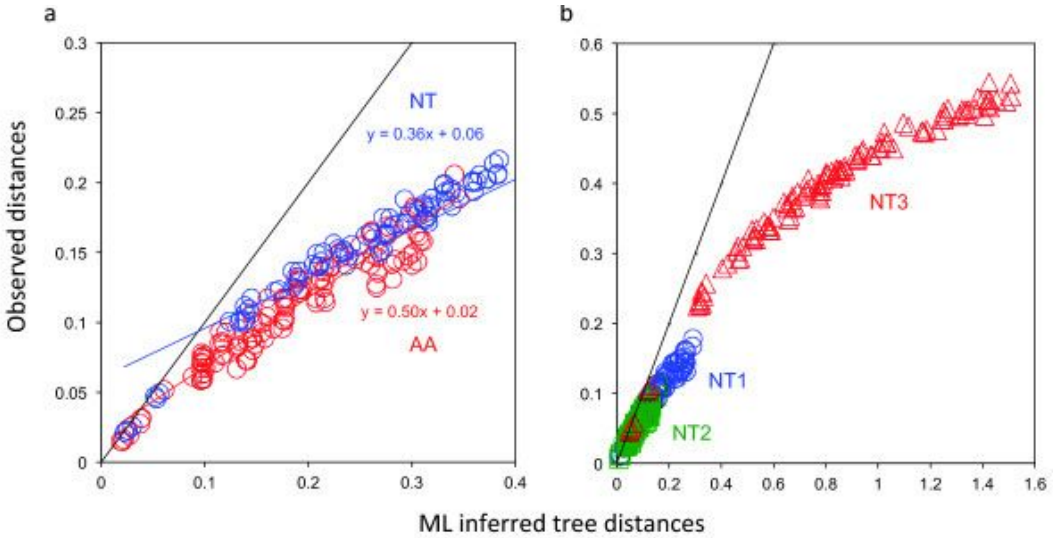
- All extant organisms had ancestors
- The relationships among all extant groups can be depicted through the patterns of divergence from their shared ancestors
- This generally takes the form of a bifurcating tree
- All sites in all genomes have histories that can be described by trees
  - Recombination and horizontal gene transfer cause these trees to be have unique topologies
- Species trees describe the aggregate evolutionary process across gene trees, accounting for:
  - HGT = horizontal gene transfer
  - ILS = incomplete lineage sorting (due to admixture)



Loci for deep-time/node vs. tip reconstruction

More loci are required to resolve deep nodes because substitutions saturate

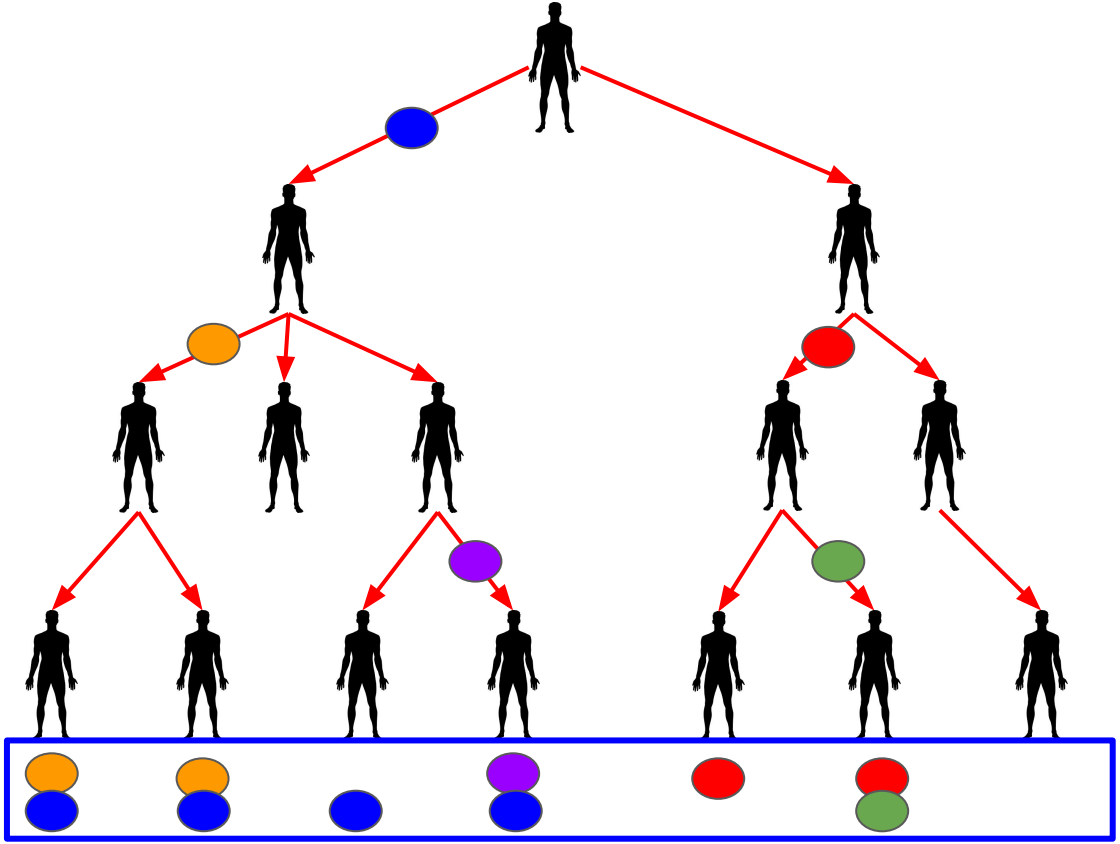




Analyses of substitution saturation at each codon position. Maximum likelihood saturation plots [34] were compared (a) between the complete amino-acid and nucleotide datasets, and (b) between the codon positions of the complete nucleotide dataset. The observed pairwise distances between the 16 taxa were directly computed from sequence alignments, and the corresponding inferred pairwise tree distances calculated from branch lengths of the ML topology. The Y = × line marks the theoretical limit where the number of observed substitutions equals the number of inferred substitutions. The slope of the linear regression indicates the amount of substitution saturation; the smaller the slope, the greater the number of inferred multiple substitutions.

Chiari et al. 2012

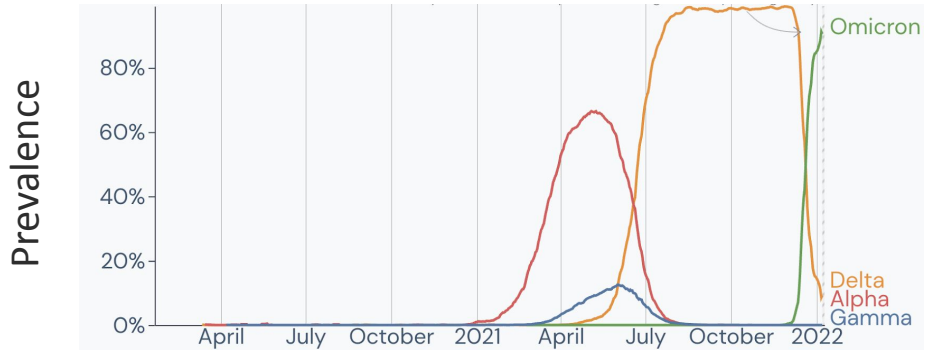
By sequencing virus genomes, we can learn about transmission and virus evolution



Slide from Russ Corbett-Detig

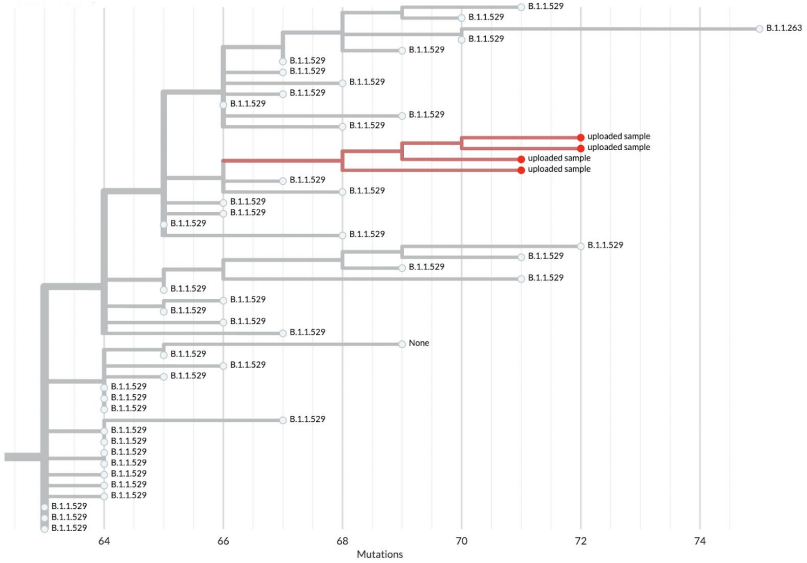
Phylogenetics Is A Central Part of the Fight Against SARS-CoV-2

1. Viral Surveillance



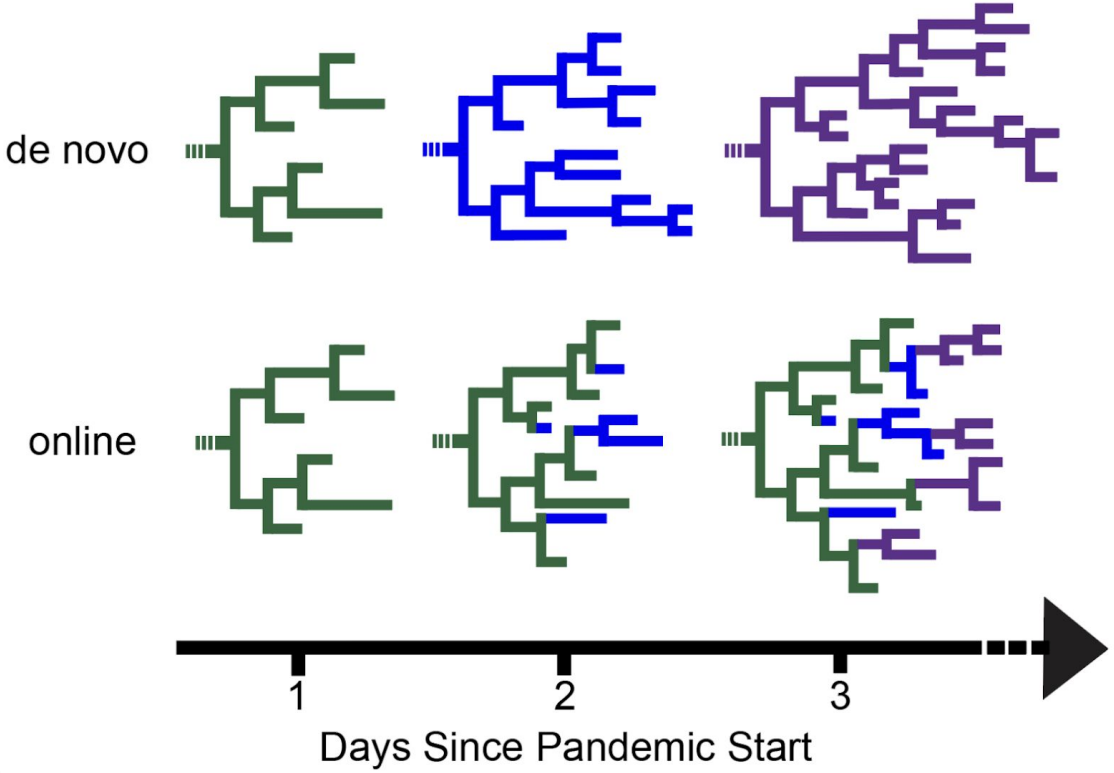
Screengrab from outbreak.info

2. Outbreak Investigation



Screengrab from UShER (on auspice)

Online Phylogenomics Is An Appealing Alternative

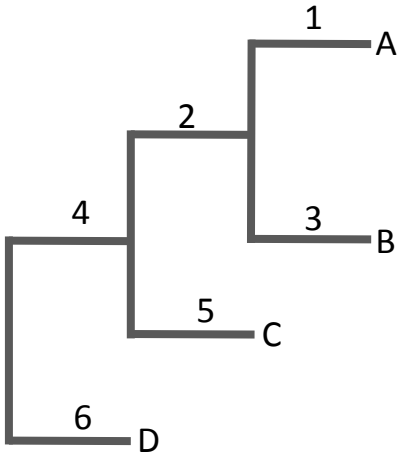


Slide from Russ Corbett-Detig

Ultrafast Sample placement on Existing tRees

UShER

UShER stores both the tree, and the inferred mutations as a **mutation annotated tree (MAT)**.



1. 6G
2. 4U
3. 7U
4. 2U
5. 9C
6. 10U

Yatish Turakhia
Now at UCSD

Turakhia et al, 2021. *Nat. Genetics.*

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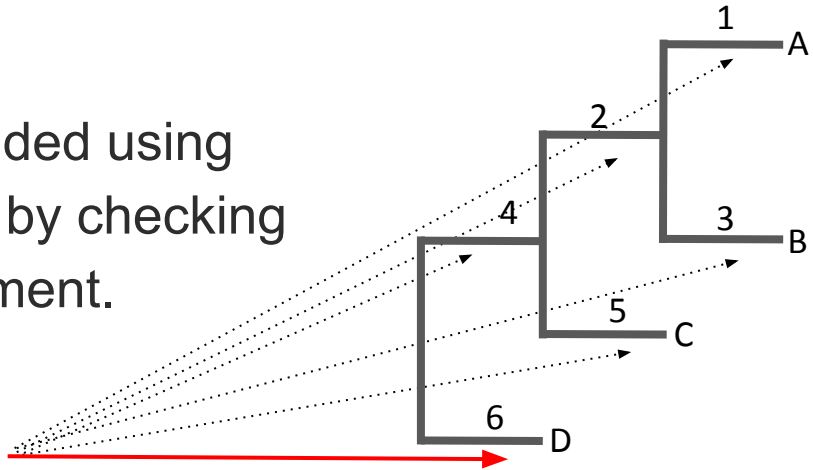
UShER Adds Samples with Brute-Force Parsimony

New samples are added using maximum parsimony by checking every possible placement.

Sample E: 10U, 14G



1. 6G
2. 4U
3. 7U
4. 2U
5. 9C
6. 10U

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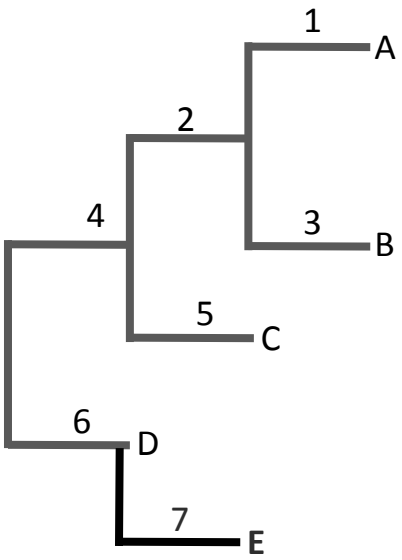
A MAT Provides Evolutionary Compression



1. 6G
2. 4U
3. 7U
4. 2U
5. 9C
6. 10U
7. 14G

A 50K sample MAT is ~1.7 Mb.

Sequence Alignment (fasta) = 1.14 Gb.

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UShER is Highly Accurate

Bryan Thornlow

Yatish Turakhia

Nicola De Maio

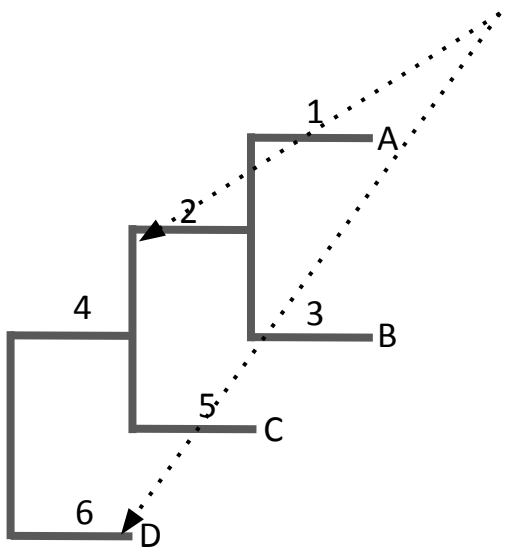
97% of placements are topologically correct.

When incorrect, typically very close.

Turakhia et al, 2021. *Nat. Genetics.*

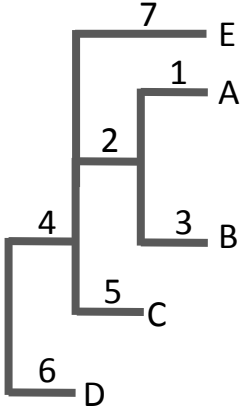
Slide from Russ Corbett-Detig

The Number of Equally Parsimonious Placements Is a Measure of Sample-Level Uncertainty for Public Health

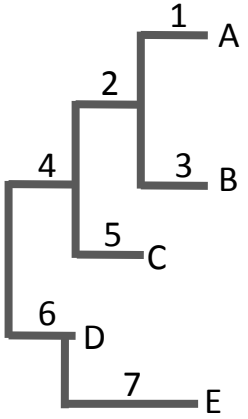


Sample E: 2U, 10U, 14U

1. 6G
2. 4U
3. 7U
4. 2U
5. 9C
6. 10U



1. 6G
2. 4U
3. 7U
4. 2U
5. 9C
6. 10U
7. 10U,14U

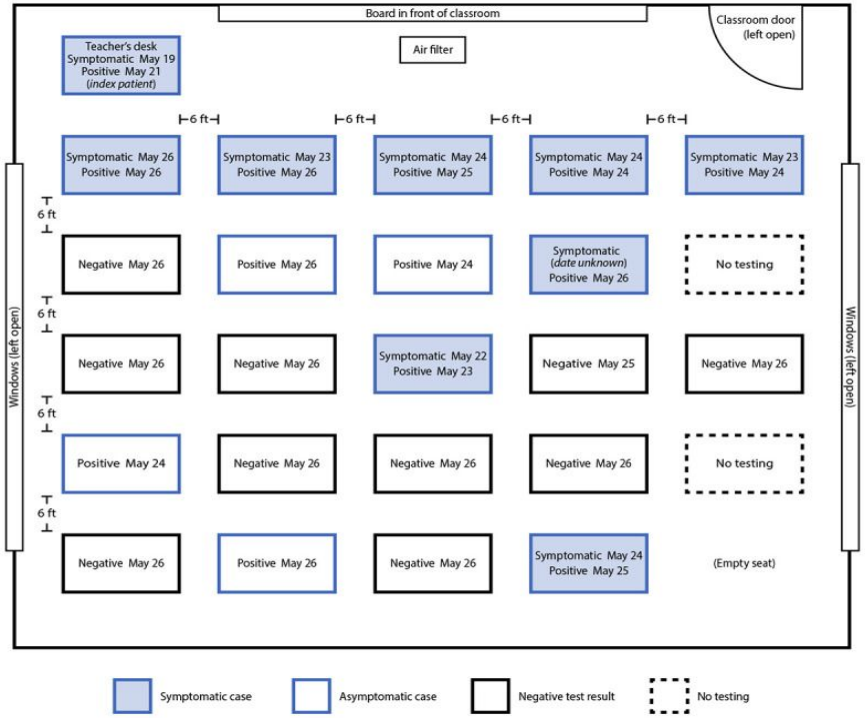


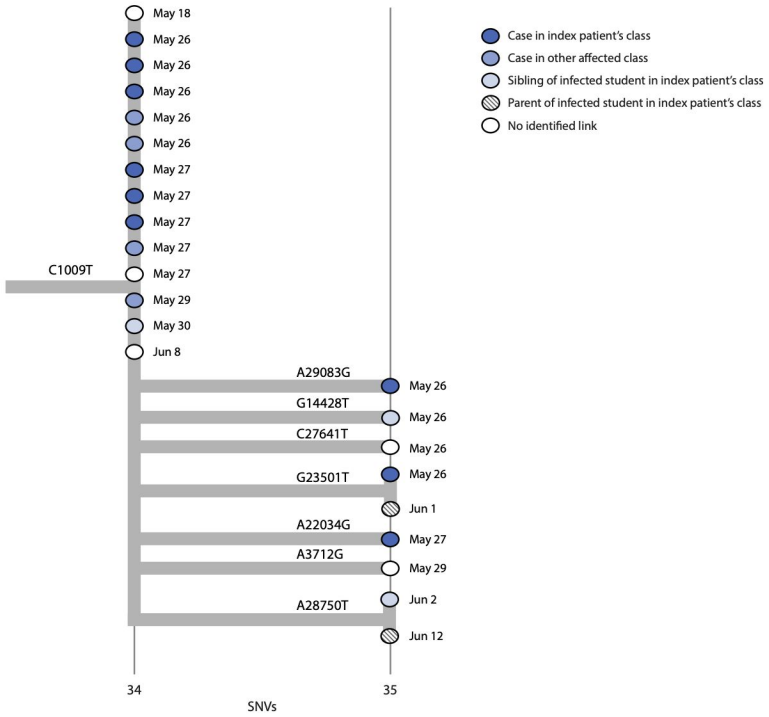
1. 6G
2. 4U
3. 7U
4. 2U
5. 9C
6. 10U
7. 2U,14U

Turakhia et al, 2021. *Nat. Genetics.*

Slide from Russ Corbett-Detig

Online Phylogenomics Powers Outbreak Investigations





Lan-Hine et al. 2021. *MMWR.*

Slide from Russ Corbett-Detig